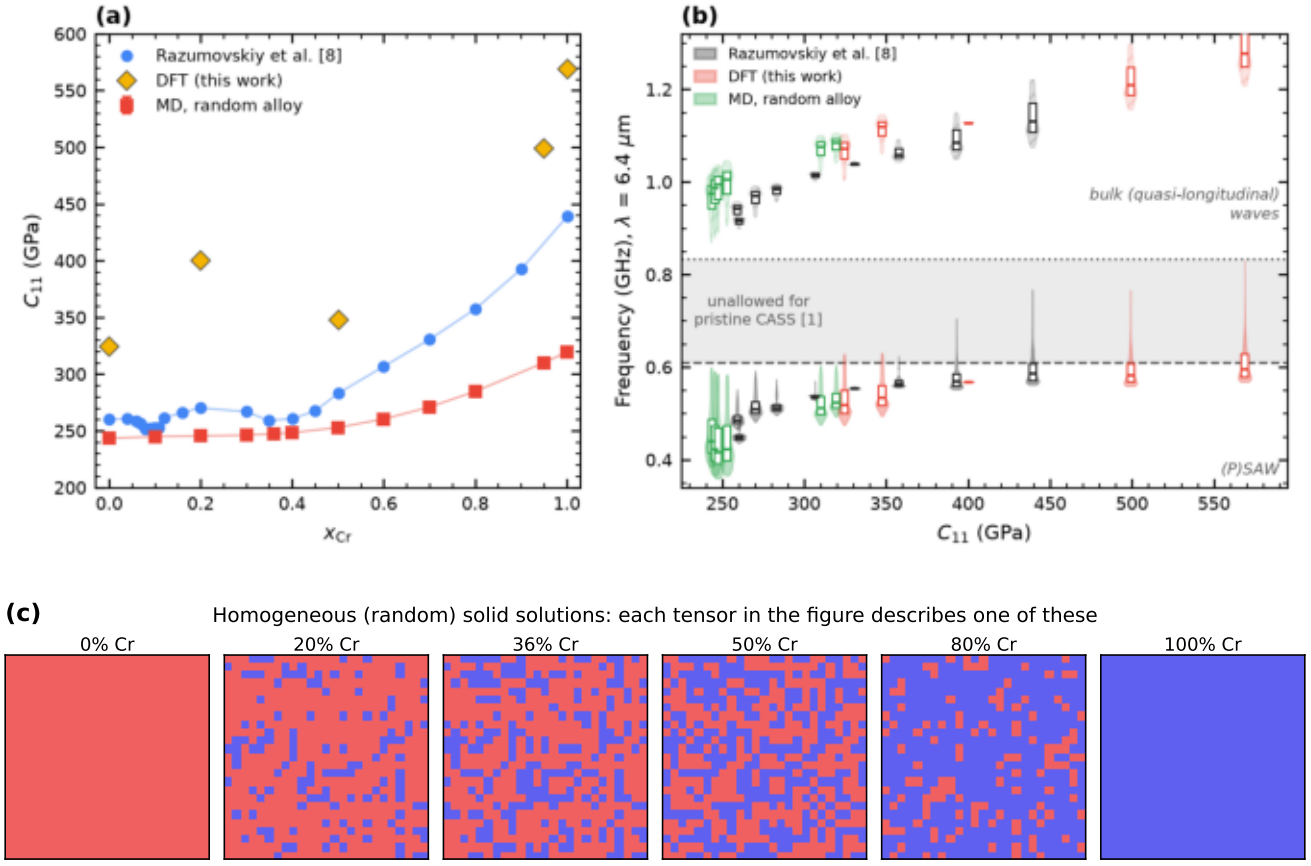

(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
C_{11} (GPa)
x_{Cr}
(b)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
Frequency (GHz), λ = 6.4 μm
bulk (quasi-longitudinal) waves
unallowed for pristine CASS [1]
(P)SAW
C_{11} (GPa)
(c)
Homogeneous (random) solid solutions: each tensor in the figure describes one of these
0% Cr
20% Cr
36% Cr
50% Cr
80% Cr
100% Cr
one atomic (001) plane of a random bcc alloy: red = Fe, blue = Cr